

Buried Resources: A Cross-Country Study of Municipal Waste Recovery

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Abstract

This study examines the gap between potentially recoverable components of municipal waste and the amount reported as recycled or biologically treated. Country-level data from What a Waste 3.0 and the World Development Indicators support a comparison of collection and post-collection gaps with those of similar countries. Of the 217 countries and economies in the source workbook, 84 meet the analysis criteria. An eight-peer benchmark yields an additional recovery proxy of 127.3 million tonnes per year across these observations. The estimate combines records from different years and does not represent measured landfill diversion.

1 Research question

Municipal waste contains materials that may be recycled or treated biologically, yet the share reaching such processes varies widely. What a Waste 3.0 reports waste generation, composition, collection, and treatment across countries and economies [1]. The central questions concern the gap between recoverable composition and reported recovery, the relative contributions of collection and subsequent treatment, and the performance of countries with similar economic and waste characteristics.

2 Data and study sample

The analysis draws on the What a Waste 3.0 country workbook and its codebook. Two World Development Indicators (WDI) describe economic development and urbanization: GDP per capita at purchasing power parity in constant 2021 international dollars (NY.GDP.PCAP.PP.KD) and urban population share (SP.URB.TOTL.IN.ZS) [2]. Each country is paired with the closest available WDI observation within three years of its reported waste-generation year. The final waste observations span 2012–2023; their sum therefore combines different reference years.

The potentially recoverable composition share, p_i , combines the mass shares of food waste, yard waste, paper and cardboard, plastics, metals, and glass. The reported recovery share, r_i , comprises recycling, composting, and anaerobic digestion. Incineration is excluded because the workbook does not consistently establish whether its output is put to productive use. The mass-based collection share, c_i , equals one minus the uncollected waste share when that measure is available; otherwise, it is the reported national share of waste mass collected. Household and population coverage have different denominators and are not substituted for waste-mass coverage.

The sample requires positive generation and a valid year, component shares within $[0, 1]$, composition shares and the combined treatment-plus-uncollected shares each totaling 95–105%, mass-based collection coverage, recovery consistent with the selected collected composition, and both WDI variables near the waste year. Table 1 records the sequential changes in sample size. Missing subcategories contribute zero only when their distribution passes the closure check.

Table 1: Sequential selection of country observations.

Criterion	Before	Retained
Source countries and economies	217	217
Positive generation and valid year	217	200
Composition shares total 95–105%	200	172
Treatment plus uncollected shares total 95–105%	172	153
Mass-based collection measure available	153	88
Recovery consistent with collected mass	88	86
Both WDI variables matched in time	86	84

3 Measures and peer comparison

Let G_i denote annual municipal waste generation in tonnes. Recovery relative to the selected composition is $e_i = r_i/p_i$, while $G_i(p_i - r_i)$ estimates the mass of selected material outside reported recovery. If uncollected waste has the same composition as the total waste stream, the proxy gap separates into two stages:

$$G_i(p_i - r_i) = \underbrace{G_i p_i (1 - c_i)}_{\text{collection gap}} + \underbrace{G_i (p_i c_i - r_i)}_{\text{post-collection gap}} . \quad (1)$$

The sample retains only records satisfying $r_i \leq p_i c_i + 0.01$, with a one-percentage-point tolerance for discrepancies in reported shares.

Peer matching uses log GDP per capita, urban population share, and p_i . Each variable is centered at the sample median and scaled by its interquartile range; Euclidean distance then identifies the eight nearest other countries. Recovery performance plays no role in the matching distance. The post-collection capture rate is $t_i = r_i/(p_i c_i)$. The 75th percentiles of peer collection and peer capture provide separate benchmarks. Targets c_i^* and t_i^* take the larger of each benchmark and the country’s current value. The additional recovery proxy is

$$\Delta_i = G_i p_i (c_i^* t_i^* - c_i t_i) = \underbrace{G_i p_i (c_i^* - c_i) t_i}_{\text{collection expansion}} + \underbrace{G_i p_i c_i^* (t_i^* - t_i)}_{\text{post-collection improvement}} . \quad (2)$$

The decomposition first raises collection and then improves capture at the expanded collection level. Because the two percentile targets may come from different peers, they describe a combined benchmark scenario rather than an observed national system.

4 Results

The 84 selected observations report 1.353 billion tonnes of annual waste generation in total, although their reference years differ. In Figure 1, several countries combine a large selected composition share with a low reported recovery-to-composition ratio. Collection coverage also varies among countries with similar composition shares, supporting a separate examination of collection and subsequent conversion.

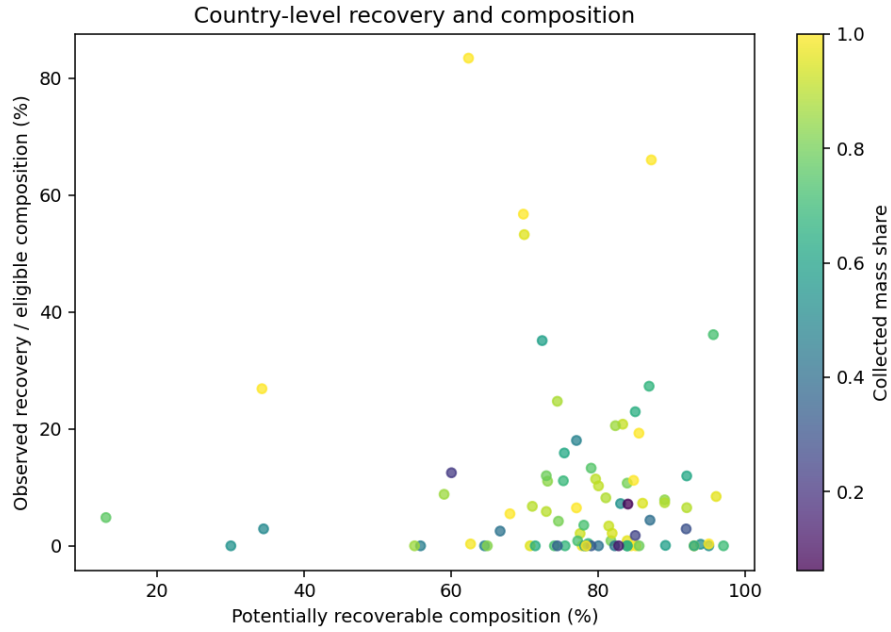


Figure 1: Selected potentially recoverable composition and observed recovery efficiency in the final sample. Color denotes the collected share of waste mass.

The eight-peer scenario yields 127.3 million additional tonnes per year across the mixed-year sample. Collection expansion contributes more than post-collection improvement in 21 countries; post-collection improvement contributes more in 55, and eight have no positive benchmark gap. Figure 2 displays the 15 largest absolute gaps. China's estimated 66.5 million tonnes illustrate how national waste volume, along with the proportional gap, shapes a ranking by tonnes.

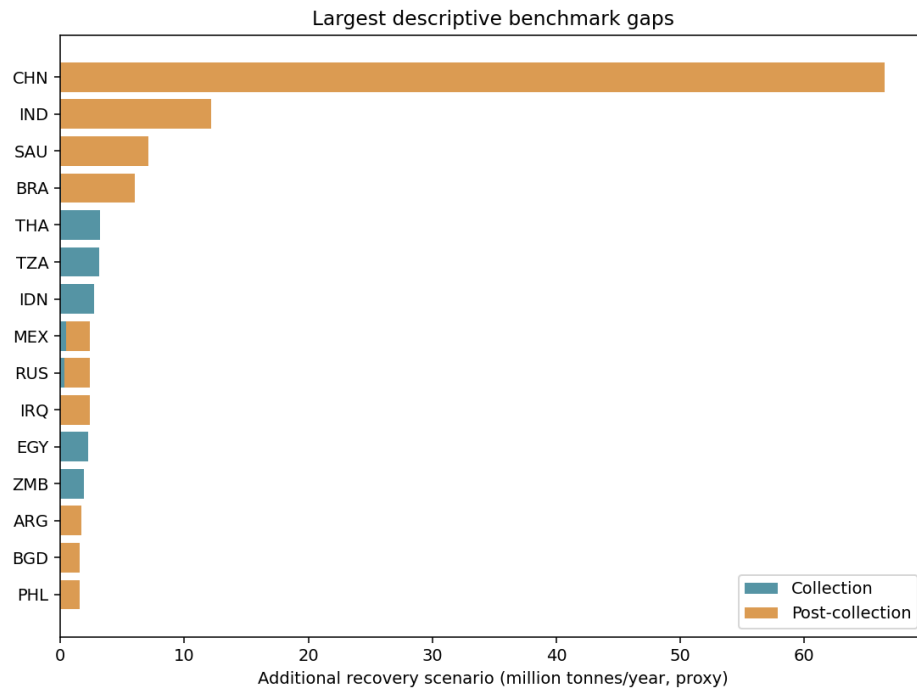


Figure 2: The 15 largest additional recovery proxies under the eight-peer benchmark. Bars separate collection expansion from post-collection improvement.

Figure 3 shows that 45 of the 84 countries have waste reference years of 2021 or 2022, whereas several observations date to 2012–2018. This spread reinforces the interpretation of the country sum as a mixed-year comparison rather than a single-year global estimate.

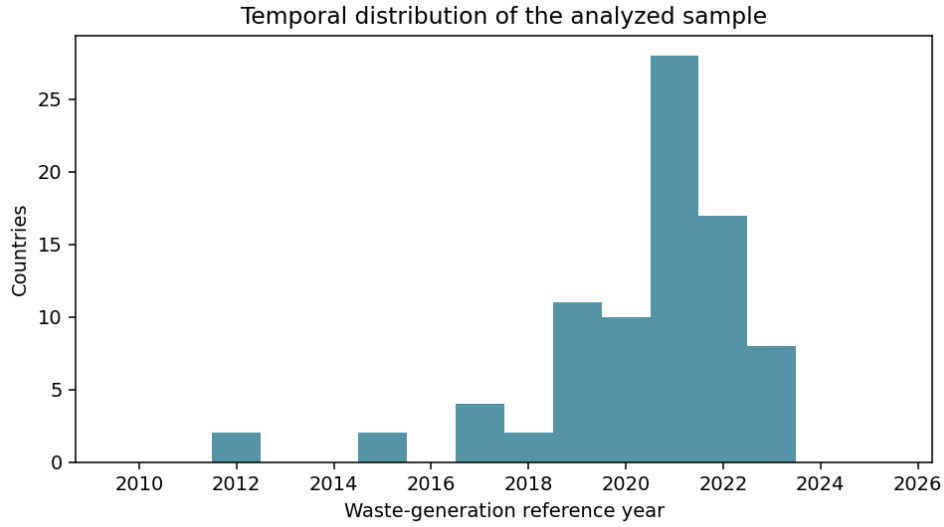


Figure 3: Waste-generation reference years for the 84 analyzed countries and economies.

Figure 4 shows that the estimated total changes non-monotonically with the number of peers. The five-, eight-, and twelve-peer settings yield 154.9, 127.3, and 141.3 million tonnes per year, respectively. All settings use the same 84 countries; the change comes from recalculating the peer percentiles. The increase from eight to twelve peers follows a decrease from five to eight, indicating that the aggregate estimate depends on the matching choice.

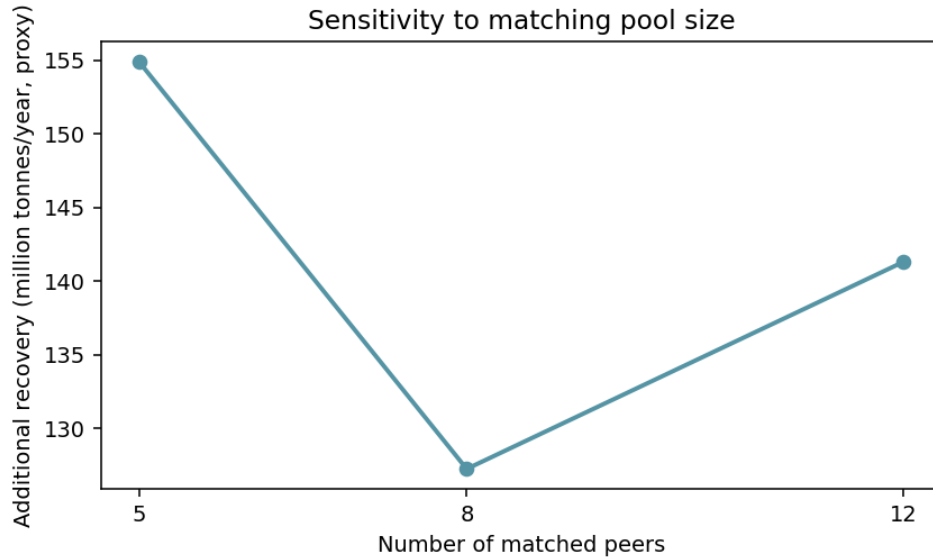


Figure 4: Additional recovery proxy under three peer-pool sizes. Each setting uses the same 84-country sample.

5 Conclusion

Across the 84 countries in the analytical sample, the gap between recoverable waste composition and reported recovery varies substantially. The two-stage analysis shows whether each gap is driven mainly by limited collection or by low recovery after collection. Comparisons with similar countries provide a benchmark, though the aggregate estimate depends on the number of peers used. The results can help identify waste systems for closer study, but they do not establish realized material flows or causal effects.

References

- [1] World Bank Group. *What a Waste 3.0: Global Snapshot of Solid Waste Management toward Circularity until 2050*, Country dataset and codebook, 2026. <https://datacatalog.worldbank.org/search/dataset/0039597/what-a-waste-global-database>.
- [2] World Bank. *World Development Indicators*, indicators NY.GDP.PCAP.PP.KD and SP.URB.TOTL.IN.ZS. <https://api.worldbank.org/v2/>.